Shounak Nandi

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Profile

I am a Biomedical engineer specializing in diffusion MRI microstructure modeling and machine learning for clinical neuroimaging, with three years of computational research experience developing HPC-based analysis pipelines for pediatric neuro-oncology cohorts. I have contributed to several peer-reviewed publications and international conference abstracts. My skill set includes quantitative MRI research, data analysis, and hands-on medical device engineering in live surgical environments, including low-field MR hardware development.

Education

Master of Science in Biomedical Engineering and Medical Physics — Apr. '21 – Apr. '24
Technical University of Munich, GPA: 3.23/4.00 — *Munich, Germany*

Bachelor of Technology in Biomedical Engineering — Aug. '13 – Jun. '17
Maulana Abul Kalam Azad University of Technology, GPA: 3.98/4.00 — *Kalyani, India*

Journal Publications

Published *co-first authors

- **Nandi S**, et al., " Magnetic Resonance Imaging Textural Changes Are More Sensitive Than Volumetric Changes in the Amygdala of Cocaine Use Disorder Patients," NMR in Biomedicine 39, no. 1 (2026): e70193, https://doi.org/10.1002/nbm.70193.
- McKenna F*, **Nandi S***, et al. Microvascular and microstructural brain abnormalities in paediatric haematological cancer survivors are related to cognitive deficits: An IVIM-FWI MRI study. Br J Haematol. 2026; 00: 1–12. https://doi.org/10.1111/bjh.70300
- Nguyen HQ, Hadidchi R, Eligulashvili A, **Nandi S**, et al. COVID-19 and radiological progression of multiple sclerosis. Diagnostics, 16(10), 1513. https://doi.org/10.3390/diagnostics16101513
- Henry SS, **Nandi S**, et al. Temporal lobe cortical thickness and cognition in paediatric cancer survivors previously treated with chemotherapy. British Journal of Haematology, 2026; 00: 1–9. https://doi.org/10.1111/bjh.70911

Accepted / In Press

- Nayak S, **Nandi S**, et al. Cognitive Control in Pediatric Cancer Survivors: A task-fMRI study; Sci Rep. 2026

Under Review

- **Nandi S***, McKenna F*, Henry S, Nayak S, Duong TQ. White matter–cognition mapping in pediatric cancer survivors: a two-cohort NODDI-Bingham study. Hum Brain Mapp (under review). Preprint: bioRxiv 2026.09.25.754456. https://doi.org/10.64898/2026.09.25.754456
- Muir ER*, **Nandi S***, et al. Neuroimaging Correlates of Chemotherapy-Related Cognitive Impairment in Pediatric Hematologic Cancer Survivors: A Systematic Review; Pediatr Blood Cancer. 2026.
- McKenna F, Hemmerling KJ, **Nandi S**, Molholm S, Zhu DC. Multimodal MRI clustering reveals an elevated free-water biotype in autism with increased social symptoms. Mol Autism (under review). Preprint: bioRxiv 2026.09.25.754196. https://doi.org/10.64898/2026.09.25.754196

Conference Abstracts & Presentations

International

- **Nandi S**, et al. Textural changes in the amygdala are more sensitive than volumetric changes in cocaine use disorder patients undergoing therapy. *Proc. Intl. Soc. Mag. Reson. Med.*, Singapore. 2024. Poster Presentation.
- Srinivas SA, Alon L, Alipour A, **Nandi S**, Purchase A, et al. MRI4ALL: A week-long hackathon for the development of an open-source ultra-low-field MRI system. *Proc. Intl. Soc. Mag. Reson. Med.*, Singapore. 2024. Oral Presentation.

- **Nandi S**, et al. Textural changes are more sensitive than volumetry of amygdala in cocaine use disorder patients: potential for low field translation. *2nd ISMRM Workshop on Accessible MRI*, New Delhi, India. 2024. Poster Presentation.
- **Nandi S**, et al. Cerebellar–cortical white-matter disorganization is associated with cognitive impairment in pediatric cancer survivors: A NODDI-Bingham study. *SOBP Annual Meeting*, New York, USA. 2026. Poster Presentation.
- **Nandi S**, Geethanath S. LASSO-informed textural changes are more sensitive than volumetry of amygdala in cocaine use disorder patients. *Intl. Soc. Mag. Reson. Med.*, Cape Town, South Africa. 2026. Poster Presentation.
- Nayak S, **Nandi S**, et al. Task-based deactivations in a fast event-related fMRI design are better captured by deconvolution than FIR basis functions. *Intl. Soc. Mag. Reson. Med.*, Cape Town, South Africa. 2026. Power-Pitch Presentation.
- McKenna FF, **Nandi S**, et al. Altered IVIM-FWI metrics in pediatric hematologic cancer survivors and their relationship with cognitive deficits. *Intl. Soc. Mag. Reson. Med.*, Cape Town, South Africa. 2026. Poster Presentation.
- **Nandi S**, et al. NODDI-Bingham reveals loss of positive white matter–cognition coupling in adult survivors of childhood cancer. *Intl. Soc. Mag. Reson. Med.*, Vancouver, Canada. 2027. *Submitted*.
- **Nandi S**, et al. Increased estimated brain-age gap in adult survivors of childhood cancer treated with chemotherapy. *Intl. Soc. Mag. Reson. Med.*, Vancouver, Canada. 2027. *Submitted*.

Domestic

- **Nandi S**, et al. Advanced Microstructure Diffusion Models in Pediatric Chemo Brain. *i2i workshop.* New York, USA. 2026. Poster Presentation.
- Poojar P, **Nandi S**, et al. Spectral quality and localization in MR spectroscopy data from opioid use disorder patients. *Annual Neuroscience Retreat.* New York, USA. 2025. Poster Presentation.

Research Experience

Research Technician Jun.'24 – present

Radiology (PI: Tim Duong), ***Albert Einstein College of Medicine*** *New York City, USA*

- Perform pre- and post-processing of multi-modal magnetic resonance imaging (MRI) data, including diffusion MRI, functional MRI, arterial spin labeling, structural T1w and FLAIR images
- Utilize machine learning and neuroimaging tools (FSL, MRtrix, Freesurfer, SPM, DIPY, AFNI) to create specialized processing pipelines for data analysis
- Build pipelines for processing big-data with use of a High-Performance Compute cluster for data analysis
- Analysis, organization, and research of clinical data and cognitive batteries for multiple studies
- Collaborate with faculty researchers and mentor high-school students for research projects in the field of oncology and neurodegenerative diseases

Visiting Scholar Jan. – May. '24

Accessible MR Lab (PI: Sairam Geethanath), ***Johns Hopkins University School of Medicine*** *Baltimore, USA*

- Analyze and optimize the low-resolution MR images from very low-field scanner, volumetric and textural analysis on substance-use-disorder patients, building classifier models to predict treatment responders

Associate Researcher I Apr. – Dec. '23

Accessible MR Lab (PI: Sairam Geethanath), ***Icahn School of Medicine at Mount Sinai*** *New York City, USA*

- Analyze and optimize the low-resolution MR images from very low-field scanner, volumetric and textural analysis on substance-use-disorder patients, building classifier models to predict treatment responders
- Use segmentation algorithms (FSL, FreeSurfer, ANTS) to perform segmentation of the brain volumes
- Perform data collection and quality control of study data, localization of the brain subcortical regions during the MR scan, and further compute MR spectroscopy and data analysis for opioid-use-disorder patients (Dept. of Psychiatry, ISMMS PI: Yasmin Hurd)

Student Research Assistant Apr. – Oct. '22

Computer-Assisted Medical Procedure Lab (PI: Nassir Navab), ***Technical University of Munich*** *Munich, Germany*

- DigiBiop- Built a Python project for automatically sorting multi-modal (PET-MR) prostate DICOM images based on the type of scan and other patient demographics
- Real-time registration of pre-operative computed tomography (CT) data with inter-operative Ultrasound data- Worked on building an algorithm that calibrates the point cloud of the real-world phantom with third-party imaging software to perform real-time optical tracking, using C++ and Python
- Acquired and analyzed 6D optical tracking data measurements and calibrated the optical tracker with the wire and body phantoms using optical markers on the same software

Industry Experience

Regional Service Engineer — Jul. '18 – Apr. '21
Karl Storz Endoskope — *Kolkata, India*

Sales and Service Engineer (Trainee) — Jun. '17 – Jul. '18
Alliance Biomedica — *Kolkata, India*

Workshops and Hackathons

Delta DIY MRI Workshop II | *MRI, DIY, Troubleshooting, Programming* — Sep. '26
Johns Hopkins University — *Baltimore, USA*

- Built a 0.25 T C-shaped low-field MR scanner over a 3-day workshop, using two permanent magnets, passive shimming, and custom gradient coils. delta-diy-mri.github.io

MRI4ALL Hackathon | *MRI, Research, Troubleshooting, Programming* — Oct. '23
New York University — *New York City, USA*

- Co-built Zeugmatron Z1, a fully open-source very low-field MR scanner, over a 5-day hackathon; the system uses ∼1000 permanent magnets with active and passive shimming and built-in liquid cooling. doi.org/10.1002/jmri.29771

Technical Skills

Methods: Diffusion modeling (DTI, DKI, NODDI-Bingham), tractography, fMRI (task & resting-state), ASL, MR spectroscopy, structural morphometry, radiomics/texture analysis, machine & deep learning
Neuroimaging: FSL, FreeSurfer, MRtrix, ANTs, AFNI, SPM, DIPY, Slicer, jMRUI
Programming: Python, C++, MATLAB, R, Shell, SQL, LaTeX
Libraries: PyTorch, NumPy, SciPy, Pandas, Matplotlib, Seaborn, Nibabel, Pydicom, Mahotas
Infrastructure: Linux, HPC/Slurm, Git, Docker

Language Skills

Bengali (native), **English** (fluent), **Hindi** (native), **German** (intermediate)

Awards & Recognitions

Delta MRI Workshop Signal Scout Award	2026
Delta MRI Workshop II Travel Grant	2026
ISMRM Annual Meeting Trainee Stipend	2024
ISMRM Accessible MRI Workshop Travel Stipend	2024
TUM BEMP Master Thesis Travel Grant	2024

Mentoring & Outreach

- **Mentor**, high school and undergraduate summer students, MRRC (2023, 2024, 2026)
- **Invited Speaker**, "Introduction to the World of Endoscopy," Core Biomedical Engineering Concepts webinar series, Bee Biomed (2020)
- **Community Manager**, DEGIS International Students Community, Munich, Germany (Jan.–May 2023)

References are available upon request.